

ENVIRONMENT DIAGRAMS AND LAMBDA EXPRESSIONS Solutions

COMPUTER SCIENCE MENTORS 61A

September 16 – September 19, 2025

1 Environment Diagrams

1. Draw the environment diagram that results from running the code. What does the program print?

```
def foo(x):  
    def bar():  
        print(x)  
    return bar
```

```
x = foo(3)  
y = foo(4)  
x()  
y()
```

<https://tinyurl.com/5n75xdk5>

2. Draw the environment diagram that results from running the following code.

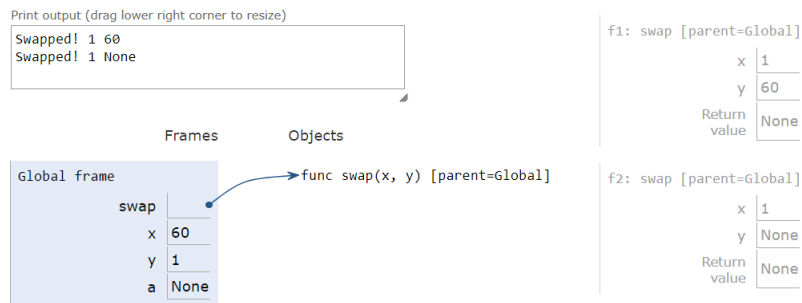
```
a = 1  
c = 2  
def b(b):  
    def d():  
        return b + c  
    return d()  
c = b(a)  
a = b(c)
```

<https://tinyurl.com/y2gj96e3>

3. Give the environment diagram and console output that result from running the following code.

```
def swap(x, y):
    x, y = y, x
    return print("Swapped!", x, y)
```

```
x, y = 60, 1
a = swap(x, y)
swap(a, y)
```



<https://tinyurl.com/y68m6qdj>

2 Lambda Expressions and Higher Order Functions

1. Express the following expressions using lambdas instead of their named counterparts.

(a) square(4)

```
(lambda x: x * x)(4)
```

(b) sum_of_squares(3, 4)

```
(lambda x, y: x * x + y * y)(3, 4)
```

(c) `def hello_world():`
`return 'Hello, world!'`
`hello_world()`

```
(lambda: 'Hello, world!')()
```

2. Make a lambda function, `make_interval()`, that takes in the upper and lower bound of an interval, and returns a function that takes in a value `x` and checks whether `x` is in the interval `[lower, upper]`, inclusive.

```
>>> make_interval = _____
>>> in_interval = make_interval(-1, 2)
>>> in_interval(0)
True
>>> in_interval(61)
False
```

```
>>> make_interval = lambda lower, upper: lambda x: x <= upper and x >=
    lower
>>> in_interval = make_interval(-1, 2)
>>> in_interval(0)
True
>>> in_interval(61)
False
```

3. Draw the environment diagram that results from running the code.

```
def a(y):
    d = 1
    b = lambda x: y(x)
    e = lambda x: x(3)
    return e(b)
d = 5
a(lambda x: 4 - x + d)
```

<https://goo.gl/9vxEwv>